Diskrete Mathematik – Woche 8 (Suggested)

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Divisors, Division with Remainders and gcd

Problem 1 ⋆

w08 1.3, 1.6, 1.7

Prove or disprove for all $a, b, c \in \mathbb{Z}$.

1. $a \mid b \wedge b \mid a \Rightarrow a = b$
2. $a \mid bc \Rightarrow a \mid b \vee a \mid c$
3. $a \mid b \Rightarrow |a| \leq |b|$

Solution:

Every statement unfolds to "there is a $k \in \mathbb{Z}$ with ..." (Def. 4.1).

1. **False**. $a = 1, b = -1$.

   What holds: $a = \pm b$. Let $b = ak$ and $a = bl$. If $b = 0$, then $a = 0$. Otherwise $b = bkl$ gives $kl = 1$, so $k = \pm 1$.
2. **False**. $a = 6, b = 2, c = 3$. True if a is prime (Lemma 4.7).
3. **False**. $a = 5, b = 0$.

   Repair: $b \neq 0$. Then $b = ak$ with $k \neq 0$, so $|b| = |a||k| \geq |a|$.

Problem 2 ⋆

w08 2.1 (negative cases), 2.4

1. Determine q and r from Theorem 4.1 for $(a, d) \in \{(-17, 5), (17, -5), (-17, -5)\}$.
2. Prove or disprove for all $a \in \mathbb{Z}$ and $d \geq 1$: $R_d(-a) = d - R_d(a)$.

Solution:

The remainder is always in $\{0, \ldots, |d| - 1\}$, whatever the signs of a and d.

1.

(a, d)	$(-17, 5)$	$(17, -5)$	$(-17, -5)$
q	-4	-3	4
r	3	2	3

   E.g. $-17 = 5 \cdot (-4) + 3$, not $5 \cdot (-3) + (-2)$, since $r \geq 0$ is required.
2. **False**. $a = d$: $R_d(-d) = 0$, but $d - R_d(d) = d$.

   It holds iff $d \nmid a$. If $d \mid a$, then $R_d(-a) = 0 \neq d = d - R_d(a)$. Otherwise let $a = dq + r$ with $0 < r < d$. Then $-a = d(-q - 1) + (d - r)$ with $0 < d - r < d$.

Problem 3 ⋆ ⋆

w08 4.1, 4.3

Let $a, b, k \in \mathbb{Z}$ with a, b not both 0, and $n \in \mathbb{N}$.

1. Prove: $\gcd(a + kb, b) = \gcd(a, b)$.
2. Determine $\gcd(n^2 + 1, n + 1)$.

Solution:

1. We show that both pairs have the same common divisors; then their gcds agree. (Also, $a + kb$ and b are not both 0.)
   - If $a = cs$ and $b = ct$, then $a + kb = c(s + kt)$.
   - If $a + kb = cs$ and $b = ct$, then $a = (a + kb) - kb = c(s - kt)$.
2. 2 if n is odd, 1 if n is even.

$$\begin{aligned}\gcd(n^2+1, n+1) &= \gcd(n^2+1-(n-1)(n+1), n+1) \quad (\text{by } 1,\ \ k=-(n-1)) \\ &= \gcd(2, n+1)\end{aligned}$$

Primes and Factorization

Problem 4 ⋆ w08 7.5

Find $u, v \in \mathbb{Z}$ with $56u + 15v = 1$.

Solution:

$u = -4$, $v = 15$. Euclid's algorithm: $56 = 3 \cdot 15 + 11$, $15 = 11 + 4$, $11 = 2 \cdot 4 + 3$, $4 = 3 + 1$. Substituting backwards:

$$\begin{aligned}1 &= 4 - 3 = 4 - (11 - 2 \cdot 4) = 3 \cdot 4 - 11 \\ &= 3(15 - 11) - 11 = 3 \cdot 15 - 4 \cdot 11 \\ &= 3 \cdot 15 - 4(56 - 3 \cdot 15) = -4 \cdot 56 + 15 \cdot 15\end{aligned}$$

Problem 5 ⋆ ⋆ w08 8.3

Prove or disprove for all $a, b, c \in \mathbb{Z}$: $a \mid c \wedge b \mid c \Rightarrow ab \mid c$. If the claim is false, repair it with an additional hypothesis.

Solution:

False. $a = b = c = 2$.

Repair: $\gcd(a, b) = 1$. By Corollary 4.5, $ua + vb = 1$ for some u, v. With $c = ak$ and $c = bl$:

$$c = uac + vbc = uabl + vbak = ab(ul + vk).$$

Problem 6 ⋆ ⋆ w08 9.3

Prove: if $2^n - 1$ is prime ($n \geq 1$), then n is prime. Is the converse true?

Solution:

To show that N is composite, write $N = xy$ and check $1 < x, y < N$, which is the step people forget.

Contraposition. $n = 1$: $2^1 - 1 = 1$ is not prime. Otherwise, let $n = ab$ with $1 < a, b < n$, and $x = 2^a$:

$$2^n - 1 = x^b - 1 = (x-1)\big(x^{b-1} + \ldots + x + 1\big),$$

with $1 < x - 1 = 2^a - 1 < 2^n - 1$.

The converse is **false**: 11 is prime, but $2^{11} - 1 = 2047 = 23 \cdot 89$.

Problem 7 ⋆ ⋆

w08 11.2

Prove: there are infinitely many primes p with $R_4(p) = 3$. Why does the same argument fail for $R_4(p) = 1$?

Solution:

Suppose $p_1, \ldots, p_k$ are all primes with remainder 3 modulo 4, and let $N = 4p_1 \cdots p_k - 1 \geq 11$.

- N is odd, so every prime divisor of N has remainder 1 or 3 modulo 4.
- $(4s+1)(4t+1) = 4(4st + s + t) + 1$, so a product of numbers with remainder 1 has remainder 1. As $R_4(N) = 3$, some prime divisor q of N has $R_4(q) = 3$.
- $q = p_i$ is impossible: then $q \mid 4p_1 \cdots p_k - N = 1$.

For remainder 1, the analogous $N = 4p_1 \cdots p_k + 1$ may have only prime divisors with remainder 3, e.g. $21 = 3 \cdot 7$.

Congruences and Modular Arithmetic

Problem 8 ⋆

w08 12.1

Prove or disprove for all $a, b, c \in \mathbb{Z}$ and $m \geq 1$: $ac \underset{m}{\equiv} bc \wedge c \underset{m}{\not\equiv} 0 \Rightarrow a \underset{m}{\equiv} b$. If the claim is false, repair it with an additional hypothesis on c.

Solution:

False. $m = 6$, $c = 2$, $a = 3$, $b = 0$: $6 \underset{6}{\equiv} 0$, but $3 \underset{6}{\not\equiv} 0$.

Repair: $\gcd(c, m) = 1$. Let c' be the inverse of c (Lemma 4.18). Then $a \underset{m}{\equiv} acc' \underset{m}{\equiv} bcc' \underset{m}{\equiv} b$ (Lemma 4.14).

Problem 9 ⋆

w08 13.3

Compute $R_{100}(3^{2026})$.

Solution:

$29.\ 3^5 = 243 \underset{100}{\equiv} 43,\ 3^{10} \underset{100}{\equiv} 43^2 = 1849 \underset{100}{\equiv} 49,\ 3^{20} \underset{100}{\equiv} 49^2 = 2401 \underset{100}{\equiv} 1$. So $3^{2026} = \left(3^{20}\right)^{101} \cdot 3^6 \underset{100}{\equiv} 729 \underset{100}{\equiv} 29$.

Problem 10 ⋆ ⋆

w08 14.4

Prove that $x^3 + y^3 + z^3 = 2030$ has no integer solutions.

Solution:

By (4.1), an integer solution is also a solution modulo every m. By Corollary 4.17, testing $x, y, z \in \mathbb{Z}_m$ is enough. So it suffices to find one m for which the sets of possible remainders of both sides are disjoint.

$m = 9$: cubes modulo 9 are $0, 1, 8 \underset{9}{\equiv} -1$. A sum of three such values lies in $\{-3, \ldots, 3\}$, i.e. is $\underset{9}{\equiv} 0, 1, 2, 3, 6, 7, 8$. But $2030 \underset{9}{\equiv} 2 + 0 + 3 + 0 = 5$.

Problem 11 ⋆ ⋆

w08 16.4

Prove: $(p-1)! \underset{p}{\equiv} -1$ for every prime p.

Solution:

$p = 2$: $1! = 1 \underset{2}{\equiv} -1$. Let $p \geq 3$.

Every $a \in \{1, \ldots, p-1\}$ has a unique inverse in $\{1, \ldots, p-1\}$ (Lemma 4.18; the inverse is not 0).

a is its own inverse iff $a \in \{1, p-1\}$: $a^2 \underset{p}{\equiv} 1 \Leftrightarrow p \mid (a-1)(a+1) \Leftrightarrow p \mid a-1 \vee p \mid a+1$ (Lemma 4.7), i.e. $a \underset{p}{\equiv} \pm 1$.

Group the factors of $(p-1)!$ into pairs $\left\{a, a^{-1}\right\}$ with $a \neq a^{-1}$; each pair contributes $\underset{p}{\equiv} 1$. Only 1 and $p-1$ remain unpaired, so $(p-1)! \underset{p}{\equiv} 1 \cdot (p-1) \underset{p}{\equiv} -1$.

Problem 12 ⋆ ⋆

w08 17.2

Find all solutions of $x \underset{6}{\equiv} 5$, $x \underset{10}{\equiv} 3$, $x \underset{15}{\equiv} 8$ with $0 \leq x < 30$, and justify that there are no others. For which $c \in \mathbb{Z}_{15}$ does the system with $x \underset{15}{\equiv} c$ instead of $x \underset{15}{\equiv} 8$ have a solution?

Solution:

For coprime n, m: $x \underset{nm}{\equiv} a \Leftrightarrow x \underset{n}{\equiv} a \land x \underset{m}{\equiv} a$ ("$\Rightarrow$": n, m divide nm; "$\Leftarrow$": Problem 5).

The moduli are not pairwise coprime, so Theorem 4.19 does not apply directly. Split them:

- $x \underset{6}{\equiv} 5 \Leftrightarrow x \underset{2}{\equiv} 1 \land x \underset{3}{\equiv} 2$
- $x \underset{10}{\equiv} 3 \Leftrightarrow x \underset{2}{\equiv} 1 \land x \underset{5}{\equiv} 3$
- $x \underset{15}{\equiv} 8 \Leftrightarrow x \underset{3}{\equiv} 2 \land x \underset{5}{\equiv} 3$

The conditions agree, so the system is equivalent to $x \underset{2}{\equiv} 1, x \underset{3}{\equiv} 2, x \underset{5}{\equiv} 3$, i.e. $x = 23$ (unique in $\mathbb{Z}_{30}$ by Theorem 4.19).

For $x \underset{15}{\equiv} c$, the system is solvable iff $c \underset{3}{\equiv} 2$ and $c \underset{5}{\equiv} 3$ (otherwise it contains contradictory congruences modulo 3 or 5), i.e. only for $c = 8$.

Problem 13 ⋆ ⋆

w08 18.1, 18.2

Alice and Bob run the Diffie-Hellman protocol (Figure 4.1) with $p = 23$ and $g = 5$.

1. Alice chooses $x_A = 6$ and Bob chooses $x_B = 15$. Compute y_A, y_B, k_{AB}, k_{BA}.
2. Eve sees only y_A and y_B. Compute k_{AB} the way Eve would for such a small p.

Solution:

1. $5^2 = 25 \underset{23}{\equiv} 2$.
   - $y_A = R_{23}(5^6) = R_{23}(2^3) = 8$.
   - $y_B = R_{23}(5^{15}) = R_{23}(2^7 \cdot 5) = R_{23}(640) = 19$.
   - $k_{AB} = R_{23}(19^6) = R_{23}((-4)^6) = R_{23}(4096) = 2$.
   - $k_{BA} = R_{23}(8^{15}) = R_{23}(2^{45}) = 2$, since $2^{11} = 2048 = 89 \cdot 23 + 1 \underset{23}{\equiv} 1$ and $2^{45} = \left(2^{11}\right)^4 \cdot 2$.
2. Eve lists 5^x modulo 23 for $x = 1, 2, \ldots$: $5, 2, 10, 4, 20, 8$. So $5^6 \underset{23}{\equiv} y_A$, and $k_{AB} = R_{23}\left(y_B^6\right) = 2$. Eve may use *any* x with $g^x \underset{p}{\equiv} y_A$: then $y_B^x \underset{p}{\equiv} g^{x_B x} \underset{p}{\equiv} y_A^{x_B} \underset{p}{\equiv} k_{BA}$. For p with 2048 bits, this list is infeasibly long.